A Hierarchical Multi-Vehicle Coordinated Motion Planning Method Based on Interactive Spatio-Temporal Corridors

Xiang Zhang, Boyang Wang, Yaomin Lu, Haiou Liu, Jianwei Gong, and Huiyan Chen

***Abstract*—Multi-vehicle coordinated motion planning has always been challenged to safely and efficiently resolve conflicts under non-holonomic dynamic constraints. Constructing spatial-temporal corridors for multi-vehicle can decouple the high-dimensional conflicts and further reduce the difficulty of obtaining feasible trajectories. Therefore, this article proposes a novel hierarchical multi-vehicle coordinated motion planning method based on interactive spatio-temporal corridors (ISTCs). In the first layer, based on the initial guidance trajectories, Mixed Integer Quadratic Programming is designed to construct ISTCs capable of resolving conflicts in generic multi-vehicle scenarios. And then in the second layer, Non-Linear Programming is settled to generate in-corridor trajectories that satisfy the vehicle dynamics. By introducing ISTCs, the multi-vehicle coordinated motion planning problem is able to be decoupled into single-vehicle trajectory optimization problems, which greatly decentralizes the computational pressure and has great potential for real-world applications. Besides, the proposed method searches for feasible solutions in the 3-D(x, y, t) configuration space, preserving more possibilities than the traditional velocity-path decoupling method. Simulated experiments in unsignalized intersection and challenging dense scenarios have been conducted to verify the feasibility and adaptability of the proposed framework.**

***Index Terms*—Multi-vehicle coordination, motion planning, intelligent vehicles, optimization method.**

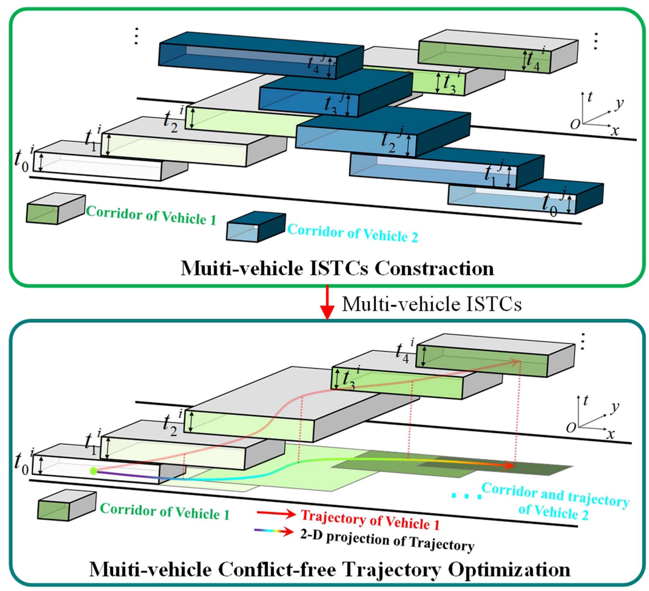


Fig. 1. Flowchart of the proposed hierarchical method.

I. Introduction

MULTI-VEHICLE collaboration technology is attracting more and more attention due to its potential value in traffic, port, warehouse and other scenarios. The rapid development of vehicle-to-everything (V2X) technology has made it possible for vehicles to interact with information from surrounding vehicles, infrastructures and command terminals [1], [2]. On this basis, the multi-vehicle planning problem can no longer be solved independently when the state and intent of each vehicle are shared. Command terminals can globally coordinate the behavior of multiple vehicles and then send instructions to corresponding individuals.

The greatest challenge for multi-vehicle coordinated motion planning is how to ensure no-conflicts for all vehicles in space-time configuration space. The current research on multi-vehicle planning mainly focuses on specific scenarios [3], such as merging and intersections. A universal method to ensure the safety and efficiency of multi-vehicle driving in common scenarios is still in deficiency. Dedicated to changing this situation, this article proposes a hierarchical planning method based on interactive spatio-temporal corridors (ISTCs), which provide a unified representation of vehicles and other environmental elements and ensures safety while efficiently avoiding conflicts.

As shown in Fig. 1, a novel hierarchical framework is proposed to accomplish the multi-vehicle coordinated motion planning in general scenes. In the first layer of the main framework, Mixed Integer Quadratic Programming (MIQP) is designed to construct ISTCs capable of resolving conflicts in generic multi-vehicle scenarios. ISTCs consist of spatial temporal corridors of all vehicles that do not overlap with each other, and the corridor

Manuscript received 26 March 2023; revised 10 May 2023; accepted 20 May 2023. Date of publication 29 May 2023; date of current version 23 February 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 52172378, and in part by the Beijing Institute of Technology Research Fund Program for Young Scholars. *(Corresponding author: Boyang Wang.)*

The authors are with the School of Mechanical Engineering, Beijing Institute of Technology, Beijing 100081, China (e-mail: xiangzhang0825@163.com; boyang_wang@bit.edu.cn; luyaomin9612@163.com; bit_lho@bit.edu.cn; gongjianwei@bit.edu.cn; chen_h_y@263.net).

Color versions of one or more figures in this article are available at https://doi.org/10.1109/TIV.2023.3280898.

Digital Object Identifier 10.1109/TIV.2023.3280898

of each vehicle is composed of corridor-cubes on a sequence of consecutive time units. Then, for the second layer, Non-Linear Programming (NLP) is settled to generate in-corridor trajectories that satisfy the vehicle dynamics.

The main contributions of the article are as follows:

- A novel hierarchical framework for handling multi-vehicle coordinated motion planning is proposed. This framework can decouple the multi-vehicle coordinated motion planning problem into single-vehicle trajectory optimization problems under specified constraints, reducing the overall complexity and improving the efficiency in general scenarios.
- An interactive spatio-temporal coupling corridor is established to achieve conflict-free zoning of each vehicle in the spatio-temporal domain. The strong non-convex conflict resolution problem in multi-vehicle scenarios can be solved effectively by setting appropriate constraints and optimization objectives in MIQP.
- Based on the generated interactive spatio-temporal corridor, each vehicle independently and optimally generates the desired trajectory in the decoupled spatio-temporal domain. Optimal trajectory generation takes into account corridor boundary constraints and vehicle motion characteristic constraints, and generates high quality trajectory timing points in a conflict decoupled space.

The remainder of this article is organized as follows. Section II presents the literature review and related works. Section III describes the framework of the proposed algorithm. Section IV describes the construction of ISTCs and the final trajectory generation. Section V demonstrates and discusses the simulation results of the ISTCs as well as the final trajectory. Finally, the conclusion and future work are given in Section VI.

II. Related Work

In this section, we review the current research status of coordinated motion planning in the fields of both robots and vehicles and present the shortcomings of current methods. In addition, an overview of the application of the spatial temporal corridor in trajectory optimization is also provided.

A. Multi-Agent Path Finding for Mobile Robotics

How to find collision-free trajectories for multi-robot systems is considered as Multi-Agent Path Finding (MAPF) problem and be widely researched for many years. As an NP-hard problem [4], MAPF is difficult to be completed efficiently and feasibly. Decoupling methods tend to achieve the shortest computation time, and they usually choose one agent to plan the path every time, ensuring that the chosen agent avoids conflicts with the plans of other agents [5], [6], [7]. However, the huge loss of solution space caused by decoupling is unacceptable in real scenes. To alleviate this problem, the method based on rules [8], A* [9], [10], [11], Conflict Based Search (CBS) [12], [13], [14] are used to find paths for multiple agents simultaneously to realize asymptotically optimal solutions. In recent years, learning-based approaches have also been widely used in MAPF. For example, Reijnen et al. used Reinforcement Learning (RL) to obtain heuristic values [15], Sinkar et al. adopted distributed Deep Learning (DL) models to deal with dynamic objects [16] and Bai et al. combined the Long Short-Term Memory (LSTM) and Deep Reinforcement Learning (DRL) to accomplish safety in formation control [17]. However, although in simple task scenarios the above methods are efficient in generating multi-agent trajectories, most of them greatly simplify or even have no regard for vehicle models and rarely consider the restraints of traffic scenarios. Hence, more details about vehicle characteristics and environmental constraints should be considered if the method is applied to real vehicles on the road.

B. Multi-Vehicle Coordinated Motion Planning

Multi-vehicle coordinated motion planning has been extensively studied in various scenarios. In the intersection scene, the method based on multi-vehicle formation control [18], Semi-Stochastic Potential Fields [19], and Partially Observable Markov Decision Process (POMDP) [20] are used to realize the cooperation of vehicles. In the lane change scenario, Reinforcement Learning (RL) combined with opponent modeling network is used to achieve efficient implementation [21]; Li et al. decentralized the lane change maneuver in two stages to balance computation speed and quality [22]. For overtaking scenarios, a method based on the artificial potential field method combined with formation control [23] and a cooperative avoidance scheme based on distance estimation strategy [24] are applied to keep vehicles safe. In addition, to deal with on-ramp coordination, artificial neural network (ANN) combined with Deep Reinforcement Learning (DRL) [25], Control Barrier Functions (CBFs) with Control Lyapunov Functions (CLFs) [26], and Satisfiability (SAT) solver [27] were proposed to generate safe trajectories and tested with good results. However, most of the above methods were only designed for specific traffic scenes.

In general scenarios, the methods based on the interactive primitive tree find the optimal mode of coordination through solving the Mixed Integer Linear Programming (MILP) [28], [29]; Reinforcement Learning (RL) is used in a discrete environment to search for configuration strategies [30]; Convex Feasible Set (CFS) algorithm [31], [32] is introduced to simplify the non-convex experiment constraints. However, these methods either apply a centralized framework to generate trajectories for all vehicles or a distributed framework to generate individual trajectories independently, with the former's computation time cursed by the number of vehicles and the latter's global optimality hardly guaranteed. In contrast, our method uses a hierarchical framework that first coordinates all vehicles centrally and then generates trajectories for each vehicle distributively, improving the solving efficiency while maintaining the collaborative effect.

C. Trajectory Optimization Based on Spatial Temporal Corridors

The spatial temporal corridor is essentially a set of passable areas for a vehicle at each moment in the entire timeline. In contrast to planning methods for decoupling paths and speeds that search in the 2-D(x, y) plane, planning methods that apply spatial temporal corridors can search for solutions in a larger

3-D(x, y, t) configuration space, leading to better trajectories under most conditions. The concept of a large convex area of accessible space and the calculation method was presented in [33]. Based on this, [34] proposed a method to formulate trajectory generation as a Quadratic Program (QP) using the concept of a Safe Flight Corridor (SFC). In the area of autonomous driving, Spatio-temporal Semantic Corridor (SSC) was first proposed in [35] to help generate the spatio-temporal trajectory for the vehicle in complex urban environments. Zhang et al. proposed a hierarchical framework consisting of rough search, fine optimization and safety strip-based collision avoidance [36], using spatial temporal corridors to ensure the safety and efficiency of ego-vehicle trajectories on spatio-temporal maps. [37] defined the spatial temporal corridor as a series of voxels with spatio-temporal connectivity, and implements behavioral planning by selecting the corresponding voxels. The trapezoidal prism-shaped corridors are introduced for optimization to generate the final trajectory in [38], [39]. Morsali et al. used support vector machines (SVM) in traffic scenarios to distinguish obstacle-free regions from the environment and construct corridors, and then used convex optimization to generate vehicle trajectories [40]. In the above approaches, the structure of spatial temporal corridors is introduced in different forms and plays an important role in ensuring the safety and efficiency of vehicle planning. However, currently, the advantages of spatial temporal corridors have rarely been utilized to deal with multi-vehicle collaboration.

The introduction of spatial temporal corridors in multi-vehicle coordinated motion planning has many advantages. The unified expression of the spatial temporal corridor enhances the applicability of the framework, which no longer limits the proposed method to specific scenarios. Besides, in-corridor trajectory optimization makes it possible to fully take into account the kinematic characteristics of vehicles, conducing to higher safety, quality and efficiency.

III. Framework

The schematic diagram of the algorithm structure is illustrated in Fig. 2. The whole hierarchical approach consists of two main parts: Multi-vehicle ISTCs construction and individual trajectory optimization. In the first part, the central computer globally calculates the ISTCs, i.e. generates a secure channel through the conflict area for each vehicle, after obtaining information about the intentions of all vehicles and environmental constraints. In the second part, the vehicles generate dynamically feasible trajectories in their respective spatial temporal corridors, no longer considering interaction conflicts and obstacle collisions.

Prior to the ISTCs calculation, initial guidance trajectories serve as inputs for the intents of vehicles, with the aim of guiding the extension of all spatial temporal corridors. In practice, for structured urban environments, we use lane lines directly as guides, while in other environments, the hybrid A* [41] algorithm is used to independently find the path of each vehicle satisfying the environmental constraints. After the path is given a constant velocity to complement the time dimension, it serves as the input initial guide trajectory. Afterwards, Mixed Integer

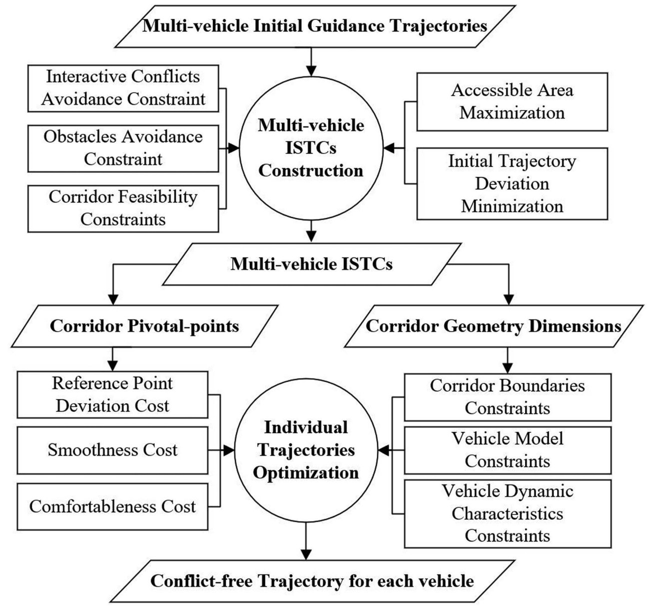


Fig. 2. Schematic diagram of the algorithm structure.

Linear Programming (MIQP) was settled to construct ISTCs. In the MIQP design, two kinds of conflicts must be considered to ensure safety: 1) Interactive conflicts among vehicles; 2) Environment conflicts between vehicles and obstacles. In addition, further types of constraints are taken into account to ensure viability: 3) Corridor feasibility constraints. All of the above three constraints can be integrated into the design of MIQP, which are detailed in Section IV. As for the objective function, since the accessible area of each vehicle is expected to be maximized and the corridor extension should be kept in line with the driving intentions, we set the optimization objectives as: 1) Maximizing the corridor-cube in each time unit; 2) Minimizing the deviation between pivotal-points of corridor-cubes and initial guidance trajectories.

In the aforementioned module, the constructed ISTCs has completely resolved the conflicts and delineated the respective passable areas for each vehicle in the space-time configuration space. Thus, the whole multi-vehicle motion planning problem can be decoupled to single-vehicle trajectory optimization problems. The generation of the final trajectories within the spatial temporal corridors is transformed into a Non-Linear Programming (NLP) problem. Several objectives are considered under multiple constraints for trajectory optimization. It is noticed that the driving intention information of the initial reference trajectories has been passed to the sequence of pivotal-points in the corridors. Therefore, for ensuring that the trajectories follow the vehicles' intents, one of the cost terms is to make generated trajectories as close to the pivotal-points as possible. Besides, the quality of trajectory is taken into account, as the cost of comfort and smoothness are indicators to be evaluated. In addition, benefiting from ISTCs, the original strongly non-convex constraint of generating conflict-free trajectories is also transformed into a linear corridor boundary constraint. Finally, vehicle model constraints and vehicle dynamic characteristic

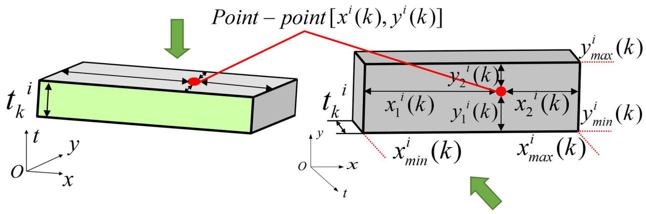


Fig. 3. Structure of a single corridor-cube.

constraints are also integrated into the NLP problem with the aim of improving the dynamical applicability of the trajectory.

In this framework, ISTCs construction is to ensure conflict-free and obstacle-free interactions, and trajectory optimization is to deliver high-quality executable trajectories. The whole method not only handles the interaction among vehicles well, but also decouples multi-vehicle motion planning into single-vehicle trajectory optimization. On this basis, the final trajectory optimization can be completed on the individual on-board computing platform with much higher real-time performance.

IV. Methodology

A. Multi-Vehicle ISTCs Construction

The initial guidance trajectories haven't taken into account conflicts among vehicles and therefore cannot be executed directly. However, the basic intent and behavior information of each vehicle provided by the guidance trajectories can be used as a reference to delineate a conflict-free passable area for each vehicle. On this basis, how to formulate the MIQP for constructing ISTCs is thoroughly described in this module.

As the conflicts between vehicles is related to their position at each moment, the spatial temporal corridor needs to be considered in both the time and space domain. The structure of the corridor is shown in Fig. 1, which represents the passable area of the vehicles on a sequence of time units. The specific manifestation of one corridor-cube is shown in Fig. 3. The position and size of the corridor-cube for vehicle i in time unit t_k^i is determined by the pivotal-point $p^i(k) = ({x_k}^i, {y_k}^i)$ and scale vector $g^i(k) = [x_1^i(k), x_2^i(k), y_1^i(k), y_2^i(k)]$. Hence, each corridor-cube can be represented by:

$$Cube^i(k) = [{x_k}^i, {y_k}^i, x_1^i(k), x_2^i(k), y_1^i(k), y_2^i(k)] \quad (1)$$

Two aspects of performance indicators mainly considered during the construction of ISTCs are listed below:

$$J_a^i = \omega_a^i \sum_k \left(x_1^i(k) + x_2^i(k) + y_1^i(k) + y_2^i(k)\right) \quad (2)$$

$$J_r^i = \omega_r^i \sum_k \left(\left(x^i(k) - x_r^i(k)\right)^2 + \left(y^i(k) - y_r^i(k)\right)^2\right) \quad (3)$$

where $(x_r^i(k), y_r^i(k))$ is the reference point provided by the initial reference trajectory, J_a^i denotes the sum of the breadth of all corridor-cubes, J_r^i indicates the sum of deviation between the pivotal-points and initial reference trajectories, and ω_a^i and ω_r^i are the corresponding weighting coefficient.

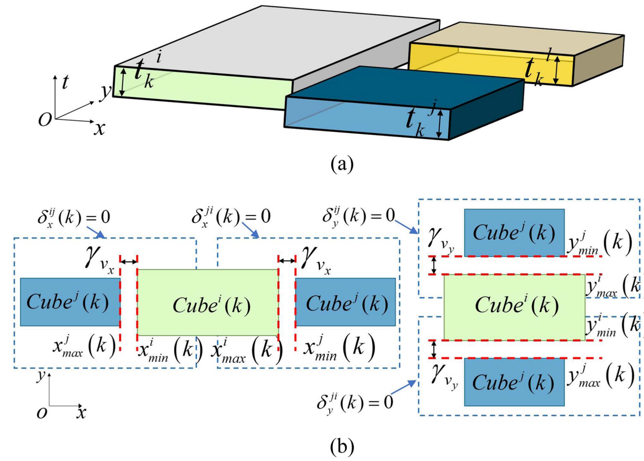


Fig. 4. (a) An example of three vehicles' corridor-cubes in a time unit. (b) The projection of corridor-cubes in $x-y$ plane when corresponding constrains is applied.

After that, the MIQP model can be formulated as follows:

$$\begin{aligned} &\underset{p^i, g^i, \forall i \in \mathcal{V}}{\text{minimize}} \sum_i \eta^i \sum_k \left(-J_a^i\left(g^i(k)\right) + J_r^i\left(p^i(k)\right)\right) \\ &\text{s.t.} \\ &\quad \forall i \in \mathcal{V}, j \in \mathcal{V}\backslash\{i\}, k \in \mathcal{K}, O_m \in \mathcal{O}: \\ &\quad Cube^i(k) \cap Cube^j(k) = \emptyset \\ &\quad Cube^i(k) \cap O_m(k) = \emptyset \\ &\quad \text{Other feasibility constraints} \end{aligned} \quad (4)$$

where the variables to be solved include the location of the pivot-points $p^i(k)$ and the the scale of the corridor-cubes $g^i(k)$, η^i is the priority weight of the vehicle i, $\mathcal{V}$ is the set of vehicles, $\mathcal{K}$ is the set of time units and $\mathcal{O}$ is the set of obstacles, $Cube^i(k) \cap Cube^j(k) = \emptyset$ is the constraint to avoid interactive conflicts among vehicles, and $Cube^i(k) \cap O^m(k) = \emptyset$ is the constraint of the environment from obstacles. Besides, in order to make sure that each corridor is dynamically feasible for its vehicle to generate a trajectory, several feasibility constraints are taken into account. In this article, the Gurobi solver is used to solve the model and the average time consumption can be maintained at the level of milliseconds.

1) $Cube^i(k) \cap Cube^j(k) = \emptyset$ (Interactive Conflicts Avoidance Constraint): This constraint is dedicated to ensuring that ISTCs is capable of keeping a safe distance between vehicles, which means that the corridor-cubes of different vehicles in the same time unit can't overlap each other. In Fig. 4(a), the corridor-cubes of three vehicles at the same time unit are shown as an example. In order to ensure noninterference, the position relationship among corridor-cubes should be considered between every two of them. Therefore, the constraint is formulated

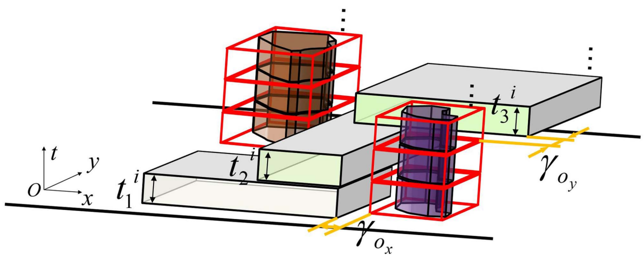


Fig. 5. An example of representing obstacles as rectangles.

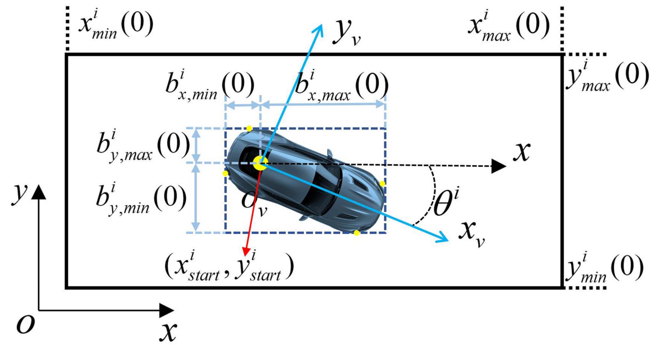


Fig. 6. $x-y$ plane projections of the car-box and corridor-cube at the initial moment.

as follows:

$$\begin{cases} x^i_{\min}(k) - x^j_{\max}(k) + \delta^{ij}_x(k)\cdot M \geq \gamma_{v_x} \\ x^j_{\min}(k) - x^i_{\max}(k) + \delta^{ji}_x(k)\cdot M \geq \gamma_{v_x} \\ y^i_{\min}(k) - y^j_{\max}(k) + \delta^{ij}_y(k)\cdot M \geq \gamma_{v_y} \\ y^j_{\min}(k) - y^i_{\max}(k) + \delta^{ji}_y(k)\cdot M \geq \gamma_{v_y} \\ \delta^{ij}_x(k) + \delta^{ji}_x(k) + \delta^{ij}_y(k) + \delta^{ji}_y(k) \leq 3 \\ \delta^{ij}_x(k) \in \{0,1\}, \delta^{ji}_x(k) \in \{0,1\}, \\ \delta^{ij}_y(k) \in \{0,1\}, \delta^{ji}_y(k) \in \{0,1\} \end{cases} \tag{5}$$

where M is the maximal number, $\delta^{ij}_x(k), \delta^{ji}_x(k), \delta^{ij}_y(k), \delta^{ji}_y(k)$ are integer variables with the value of 1 or 0, γ_{v_x} and γ_{v_y} are the safety thresholds in the x and y directions between vehicles, $[x^i_{\min}(k), x^i_{\max}(k), y^i_{\min}(k), y^i_{\max}(k)]$ and $[x^j_{\min}(k), x^j_{\max}(k), y^j_{\min}(k), y^j_{\max}(k)]$ are the boundaries of $Cube^i$ and $Cube^j$ in the same time unit. If one of the integer variables is equal to 1, means that the inequality containing this integer variable is always true so that the corresponding constraint is disabled. But if the integer variables are equal to 0, the effect is shown in Fig. 4(b). For example, when $\delta^{ij}_x(k)$ is equal to 0, the first inequality is true only if the left boundary value $x^i_{\min}(k)$ of $Cube^i$ is bigger than the right boundary value $x^i_{\min}(k)$ of $Cube^j$. In the same way, the cases of the other three integer variables equal to 0 are also shown in Fig. 4(b). At least one of the four integer variables equaling 0 ensures that in each time unit the passable areas of two vehicles are independent of each other. By adding this constraint to all vehicles in pairs, the safety of the entire interaction is ensured.

2) $Cube^i(k) \cap O_m(k) = \emptyset$ (Obstacles Avoidance Constraint): Due to spatial temporal corridor can provide a unified representation of various semantic elements in environments, we can also represent the obstacle as a cube box and use the method similar to (5) to ensure non-conflicts between vehicles and obstacles, as shown in Fig. 5. Hence, the constraints can be described as:

$$\begin{cases} x^i_{\min}(k) - x^{O_m}_{\max}(k) + \delta^{i,O_m}_x(k)\cdot M \geq \gamma_{o_x} \\ x^{O_m}_{\min}(k) - x^i_{\max}(k) + \delta^{O_m,i}_x(k)\cdot M \geq \gamma_{o_x} \\ y^i_{\min}(k) - y^{O_m}_{\max}(k) + \delta^{i,O_m}_y(k) M \geq \gamma_{o_y} \\ y^{O_m}_{\min}(k) - y^i_{\max}(k) + \delta^{O_m,i}_y(k)\cdot M \geq \gamma_{o_y} \\ \delta^{i,O_m}_x(k) + \delta^{O_m,i}_x(k) + \delta^{i,O_m}_y(k) + \delta^{O_m,i}_y(k) \leq 3 \\ \delta^{i,O_m}_x(k) \in \{0,1\}, \delta^{O_m,i}_x(k) \in \{0,1\}, \\ \delta^{i,O_m}_y(k) \in \{0,1\}, \delta^{O_m,i}_y(k) \in \{0,1\} \end{cases} \tag{6}$$

where $[x^{O_m}_{\min}(k), x^{O_m}_{\max}(k), y^{O_m}_{\min}(k), y^{O_m}_{\max}(k)]$ is the cubic boundary of obstacle O_m in time unit t^i_k, γ_{o_x} and γ_{o_y} are the safety thresholds in x and y directions, $\delta^{i,O_m}_x(k)$, $\delta^{O_m,i}_x(k)$, $\delta^{i,O_m}_y(k)$, $\delta^{O_m,i}_y(k)$ are integer variables. The effect of integer variables and the relationship between vehicle i and the cube box of the obstacle are similar to that shown in Fig. 4(b).

3) Corridor Feasibility Constraints: After ensuring the spatial temporal corridor of each vehicle is conflict-free with other vehicles and obstacles, we also need to add constraints to ensure that the corridor is executable for the vehicle to generate passable trajectories in subsequent modules. Therefore, we need to take account of vehicle characteristics when constructing ISTCs and make sure the vehicle can reach every corridor-cube at the specified time. These kinds of constraints are mainly considered in two aspects. One is the dimension and boundary constraints of every single corridor-cube in each time unit. Each corridor-cube should be able to accommodate the car-box, and due to the limitations of vehicle dynamic characteristics, the length of each corridor-cube should not exceed the maximum distance that the vehicle can drive within a unit time period. Another is the dimension and boundary constraints of adjacent corridor-cubes between continuous time periods. Considering the behavioral continuity of vehicles, the adjacent corridor-cubes need to have a minimum overlap to make sure the vehicle can transition from the current corridor-cube to the next. Meanwhile, the distance between pivotal-points of adjacent corridor-cubes should be limited according to the maximum speed of vehicles.

The dimension and boundary constraints of the corridor-cube in a time unit are firstly considered. Each corridor-cube should have enough space to encase the car-box of vehicle with uncertain orientation.

$$\begin{cases} x^i_1(k) + x^i_2(k) \geq \gamma^i_{b_x} \\ y^i_1(k) + y^i_2(k) \geq \gamma^i_{b_y} \end{cases} \tag{7}$$

where $\gamma^i_{b_x}, \gamma^i_{b_y}$ are the minimum distance thresholds along x and y axes required to accommodate the vehicle geometry in the every time period. In practical, we set $\gamma^i_{b_x} = \gamma^i_{b_y} = \sqrt{(l^i)^2 + (w^i)^2}$, where l^i and w^i are the length and width of vehicle. Besides, since the initial states of vehicles are deterministic and non-adjustable, the corresponding corridor-cube has more stringent constraints, as shown in Fig. 6. The following constraint is applied:

$$\begin{cases} x^i_{\min}(0) \le x^i_{start} + b^i_{x,\min}(0) \\ x^i_{\max}(0) \ge x^i_{start} + b^i_{x,\max}(0) \\ y^i_{\min}(0) \le y^i_{start} + b^i_{y,\min}(0) \\ y^i_{\max}(0) \ge y^i_{start} + b^i_{y,\max}(0) \end{cases} \tag{8}$$

where $(x^i_{start}, y^i_{start})$ is the initial position of the vehicle, $[x^i_{\min}(0), x^i_{\max}(0), y^i_{\min}(0), y^i_{\max}(0)]$ is the boundary of the start corridor-cube in x and y directions, $[b^i_{x,\min}(0), b^i_{x,\max}(0), b^i_{y,\min}(0), b^i_{y,\max}(0)]$ is the distance between the edge of the car-box and the position of the vehicle that can be calculated from the state and geometry of the vehicle. In addition, due to the limitations of the maximum speed and acceleration of the vehicle, the dimensions of each corridor-cube in a unit time period are limited. The driving range constraint in time unit t^i_k of vehicle i is as follows:

$$\begin{cases} x^i_{\min}(k) \ge \alpha^i_x \cdot d^i_{x,\min}(k) \\ x_{\max}(k) \le \alpha^i_x \cdot d^i_{x,\max}(k) \\ y^i_{\min}(k) \ge \alpha^i_y \cdot d^i_{y,\min}(k) \\ y^i_{\max}(k) \le \alpha^i_y \cdot d^i_{y,\max}(k) \end{cases} \tag{9}$$

where $[d^i_{x,\min}(k), d^i_{x,\max}(k), d^i_{y,\min}(k), d^i_{y,\max}(k)]$ is the minimum and maximum boundaries of vehicle driving range in the x and y directions. Besides, notice that the relaxation factor α^i_x, α^i_y is set to increase the success rate during solving the MIQP problem. Usually we set α^i_x and α^i_y equal to 2. The four driving range values can be calculated by formulation below:

$$\begin{aligned} & d^i_{x,\min}(k) \\ & = \begin{cases} x^i(k-1) + \gamma^i_{s-} \cos\theta^i_r(k-1), \text{ if } \cos\theta^i(k-1) \ge 0 \\ x^i(k-1) + \gamma^i_{s+} \cos\theta^i_r(k-1), \text{ else} \end{cases} \\ & d^i_{x,\max}(k) \\ & = \begin{cases} x^i(k-1) + \gamma^i_{s-} \cos\theta^i_r(k-1), \text{ if } \cos\theta^i(k-1) \le 0 \\ x^i(k-1) + \gamma^i_{s+} \cos\theta^i_r(k-1), \text{ else} \end{cases} \\ & d^i_{y,\min}(k) \\ & = \begin{cases} y^i(k-1) + \gamma^i_{s-} \sin\theta^i_r(k-1), \text{ if } \sin\theta^i(k-1) \ge 0 \\ y^i(k-1) + \gamma^i_{s+} \sin\theta^i_r(k-1), \text{ else} \end{cases} \\ & d^i_{y,\max}(k) \\ & = \begin{cases} y^i(k-1) + \gamma^i_{s-} \sin\theta^i_r(k-1), \text{ if } \sin\theta^i(k-1) \le 0 \\ y^i(k-1) + \gamma^i_{s+} \sin\theta^i_r(k-1), \text{ else} \end{cases} \end{aligned} \tag{10}$$

where γ^i_{s+}, γ^i_{s-} are the maximum distance thresholds of vehicle i driving forward and backward along the current heading angle, $\theta^i(k-1)$ is the heading angle in time unit t^i_{k-1} obtained according to the initial guidance trajectory. γ^i_{s+} and γ^i_{s-} directly constrain the maximum driving range in each time unit, and their values depend on the allowed maximum and minimum speed of the vehicle while performing the collision avoidance task.

The dimension and boundary constraints of adjacent corridor-cubes between continuous time periods also need to be considered. In order to ensure that vehicles' transition between successive corridor-cubes is feasible, a minimum overlap requirement for two consecutive corridor-cubes should be imposed, as reflected in Fig. 7(a). In the x-y plane projection shown in Fig. 7(b), we can more intuitively observe the overlapping relationship between the two corridor-cubes. Sufficient overlapping area between two corridors-cubes will be guaranteed as long as position relation is constrained along x and y directions respectively. In Fig. 7(c), four basic situations of overlaps are shown with the arrows pointing in the direction of increasing imbricate areas. Any other overlapping cases that meet the requirements evolved from the four basic cases. The constraints can be described as follows:

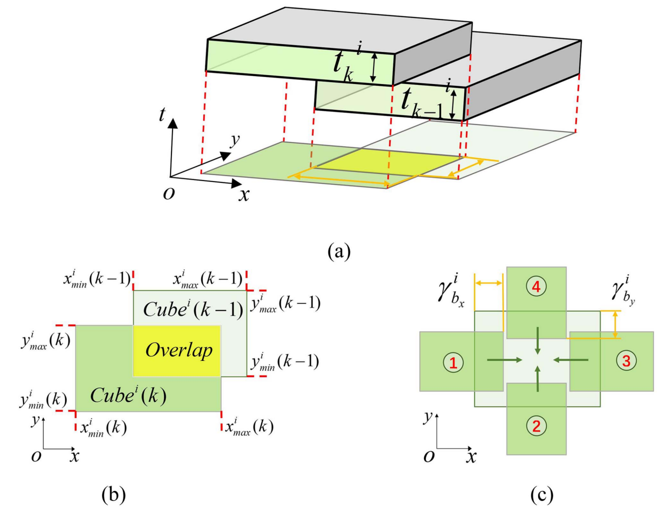


Fig. 7. (a) Overlap constraints between adjacent corridor-cubes; (b) The 2-D $x-y$ plane projection of overlap between two adjacent corridor-cubes; (c) Four basic cases of overlap between adjacent corridor-cubes.

$$\begin{cases} x^i_{\max}(k) - x^i_{\min}(k-1) \ge \gamma^i_{b_x} \\ x^i_{\min}(k) - x^i_{\max}(k-1) \le \gamma^i_{b_x} \\ y_{\max}(k) - y_{\min}(k-1) \ge \gamma^i_{b_y} \\ y^i_{\min}(k) - y^i_{\max}(k-1) \le \gamma^i_{b_y} \end{cases} \tag{11}$$

where $\gamma^i_{b_x}, \gamma^i_{b_y}$ have the same meaning as the (7). Finally, the traveling distance between the pivotal-points of the continuous corridor-cubes is limited by the maximum speed of vehicles, which can be described as follows:

$$\begin{cases} |x^i(k) - x^i(k-1)| \le |\gamma^i_{s+} \cos\theta^i_r(k-1)| \\ |y^i(k) - y^i(k-1)| \le |\gamma^i_{s+} \sin\theta^i_r(k-1)| \end{cases} \tag{12}$$

where $|x^i(k) - x^i(k-1)|$ and $|y^i(k) - y^i(k-1)|$ are the distance between two adjacent pivotal-points in the x and y directions and γ^i_{s+} has the same meaning as in (10). This formula restricts the distance between two adjacent pivotal-points smaller than the component of the maximum forward distance on both x and y axis.

B. Multi-Vehicle Conflict-Free Trajectory Generation

After the construction of ISTCs, the whole multi-vehicle planning problem is decoupled into single vehicle trajectory planning problems. Two features of spatial temporal corridor play a large role in trajectory optimization, namely the pivotal-points and corridor boundaries. The former is utilized to guide the direction of optimal trajectory, while the latter can constrain the trajectory within a safe and feasible driving area. Since each corridor is constructed in the three-dimensional space-time configuration space, we can use the lateral and longitudinal coupling method to solve the trajectory optimization problem and generate the trajectory in the spatio-temporal map directly.

In this module three costs will be considered to get the final trajectory: cost of smoothness J^i_{smooth}, cost of comfort $J^i_{comfort}$ and cost of deviation from pivotal-point $J^i_{pivotal}$. The cost of smoothness is related with the curvature κ^i_t of trajectory:

$$J^i_{smooth} = \omega^i_\kappa \sum_t \kappa^{i2}_t \tag{13}$$

where ω^i_κ is the smoothness weight of vehicle i and t represents the moment corresponding to the trajectory point. The cost of comfort consists of longitudinal and lateral parts:

$$J^i_{comfort} = \omega^i_\beta \sum_t \left(\beta^i_t\right)^2 + \omega^i_j \sum_t \left(j^i_t\right)^2 \tag{14}$$

where β^i_t is the change rate of the front wheel angle with the weight of ω^i_β, and j^i_t is the change rate of the acceleration with the weight of ω^i_j. The cost of deviation from pivotal-points is as follows:

$$\begin{aligned} J^i_{pivotal} = \omega^i_{px} &\sum_{k,t\in\left[t_k, t^i_{k+1}\right)} \left(x^i_t - x^i(k)\right)^2 \\ + \omega^i_{py} &\sum_{k,t\in\left[t_k, t^i_{k+1}\right)} \left(y^i_t - y^i(k)\right)^2 \end{aligned} \tag{15}$$

where ω^i_{px} and ω^i_{py} represent the deviation weights of the vehicle i from the pivotal-point in the x and y directions, respectively.

Then, the NLP problem can be formulated:

$$\begin{aligned} &\underset{\beta^i_t, j^i_t}{\text{minimize}} \ \left(J^i_{smooth} + J^i_{comfort} + J^i_{pivotal}\right) \\ &s.t. \\ &\quad \forall vehicle\, i,\ time\, t,\ \beta^i_t \in \mathbb{R}, j^i_t \in \mathbb{R} \\ &\quad S^i_{t+1} = f(S^i_t, \beta^i_t, j^i_t) \\ &\quad \sum_{t=0}^{t} \beta^i_t \in [\delta^i_{\min}, \delta^i_{\max}], \sum_{t=0}^{t} j^i_t \in [a^i_{\min}, a^i_{\max}] \\ &\quad (x^i_t, y^i_t) \in Corridor^i \end{aligned} \tag{16}$$

where $[\delta^i_{\min}, \delta^i_{\max}]$ is the range of front wheel steering angle, $[a^i_{\min}, a^i_{\max}]$ is the range of acceleration, S^i_t is the state parameters set of vehicle i at time t, $S^i_{t+1} = f(S^i_t, \beta^i_t, j^i_t)$ is the state transition at each timestep with the equality constraints of the vehicle model, and can be formulated as follows:

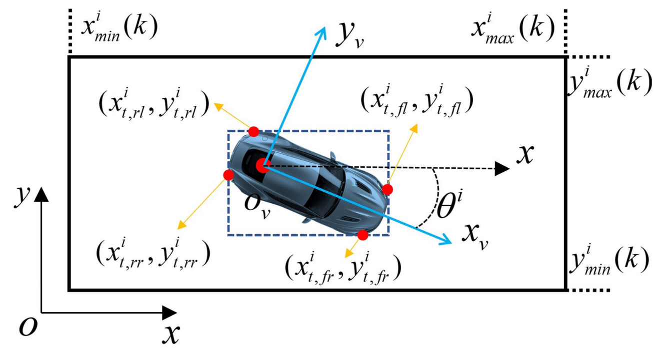


Fig. 8. Constraint to keep the vehicle inside corridor-cubes.

$$\begin{bmatrix} x^i_{t+1} \\ y^i_{t+1} \\ \theta^i_{t+1} \\ \delta^i_{t+1} \\ v^i_{t+1} \\ a^i_{t+1} \end{bmatrix} = \begin{bmatrix} x^i_t + v^i_t \cos\theta^i_t \Delta t \\ y^i_t + v^i_t \sin\theta^i_t \Delta t \\ \theta^i_t + v^i_t \frac{\tan\delta^i_t}{L} \Delta t \\ \delta^i_t \\ v^i_t + a^i_t \Delta t \\ a^i_t \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \Delta t & 0 \\ 0 & 0 \\ 0 & \Delta t \end{bmatrix} \begin{bmatrix} \beta^i_t \\ j^i_t \end{bmatrix} \tag{17}$$

the equation above is a discretized kinematic bicycle model of wheeled vehicles, which is assumed that the vehicle is driving on a relatively flat road without considering the vertical and roll motion. The vehicle dynamic characteristics constraints of the vehicle, such as the maximum acceleration and the maximum front wheel steering angle are described as:

$$\begin{cases} \delta^i_{\min} \le \delta^i_t \le \delta^i_{\max} \\ a^i_{\min} \le a^i_t \le a^i_{\max} \end{cases} \tag{18}$$

Besides, $(x^i_t, y^i_t) \in Corridor^i$ is the constraint to keep the trajectory inside the spatial temporal corridor, which means that the four corner points of the vehicle should be enclosed by corridor-cubes of its corridor at all times, as shown in Fig. 8. The constraint can be described as:

$$\begin{cases} x^i_{\min}(k) \le x^i_{t,fr} \le x^i_{\max}(k), y^i_{\min}(k) \le y^i_{t,fr} \le y^i_{\max}(k) \\ x^i_{\min}(k) \le x^i_{t,fl} \le x^i_{\max}(k), y^i_{\min}(k) \le y^i_{t,fl} \le y^i_{\max}(k) \\ x^i_{\min}(k) \le x^i_{t,rl} \le x^i_{\max}(k), y^i_{\min}(k) \le y^i_{t,rl} \le y^i_{\max}(k) \\ x^i_{\min}(k) \le x^i_{t,rr} \le x^i_{\max}(k), y^i_{\min}(k) \le y^i_{t,rr} \le y^i_{\max}(k) \end{cases} \tag{19}$$

V. Experimental Results and Discussion

In this section, we organized several groups of simulation experiments in unsignalized intersection scenario as well as in challenging dense scenarios. The proposed algorithm was implemented in C++ and executed on a laptop running Ubuntu 18.04 with Intel i7-8700H @2.30 GHz CPU and 16 GB of RAM. We used lpsolve solver [42] and Gurobi solver [43] to solve the MIQP of ISTCs constructions and used IPOPT [44] to solve the NLP for generating final trajectories inside each spatial temporal corridor. Besides, GridMap was used to represent the occupancy map of the environment. In all experiments, we discretized time in seconds to solve MIQP problem for constructing ISTCs, i.e. each corridor-cube represents a passable area in one second. The

TABLE I
RELEVANT PARAMETERS OF THE PROPOSED METHOD

Parameters	Value	Parameters	Value	Parameters	Value
γ_{v_x}	0.5(m)	γ^i_{s-}	5(m)	ω^i_j	1.0
γ_{v_y}	0.5(m)	ω^i_a	0.50	ω^i_{px}	1.00
γ_{o_x}	0.5(m)	ω^i_r	10.00	ω^i_{py}	1.00
γ_{o_y}	0.5(m)	ω^i_κ	1.00		
γ^i_{s+}	10(m)	ω^i_β	100.0		

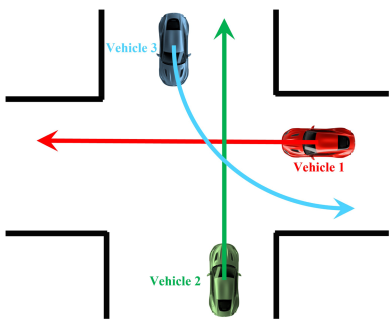


Fig. 9. Three vehicles simulated at the unsignalized intersection.

TABLE II
PRIORITY WEIGHTS OF EACH VEHICLE IN TURN-FIRST SCENARIO AND STRAIGHT-FIRST SCENARIO

Group	η^1	η^2	η^3
I	0.25	0.01	0.50
II	0.50	0.25	0.01

dimensional parameters of all vehicles were set to: length $= 4$ m, width $= 2$ m, and wheelbase $= 2.7$ m. Besides, the relevant parameter settings of the algorithm are shown in Table I.

A. Simulated Experiments of Unsigenaled Intersection

1) Description: Simulation of three vehicles from different directions in an unsignalized intersection scenario is shown in Fig. 9. $vehicle1$ and $vehicle2$ came from different directions with the intention of going straight, and $vehicle3$ came from a different direction than the first two with the intention of turning left. In this structured scenario, the initial guidance trajectories can be projected based on the vehicles' intent in combination with the lane lines. Therefore, the initial guidance trajectory of each vehicle was indicated by different colored lines in Fig. 9, where the width of the road is 10 m. Two sets of experiments are conducted as a comparison to reveal the effects of different priority weights. In the first set we let the left-turning $vehicle3$ have the higher priority, while in the second set we gave the higher priority to straight-going $vehicle1$ and $vehicle2$, as shown in Table II. In both the groups of experiments, the smallest priority weight was almost equal to 0, in order to increase the gradient of the objective function and increase the solving speed.

2) Results and Discussion: The constructed ISTCs of both groups can be generated as expected, shown in Fig. 10. Based on the initial guidance trajectories, the corridor-cubes of each vehicle are expanded in the right direction until leaving the

TABLE III
TIME CONSUMPTION OF THE WHOLE METHOD

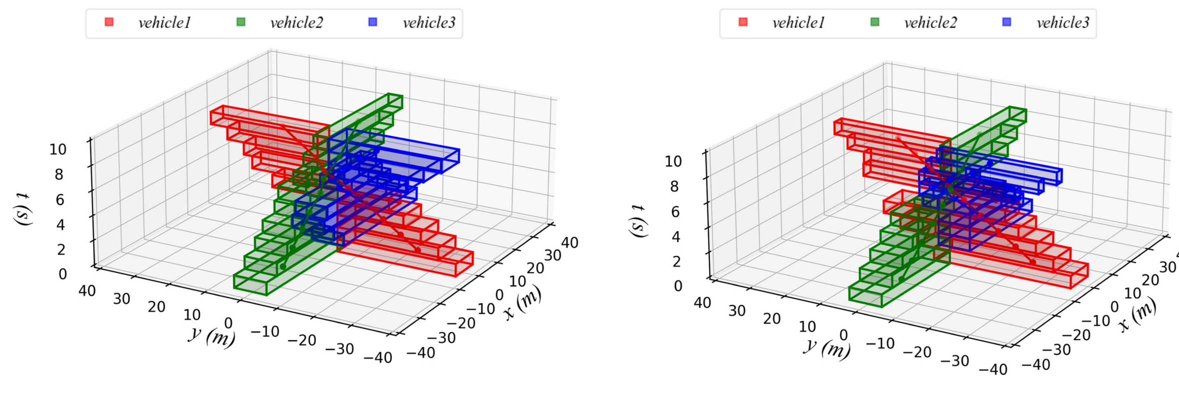
Group	t_{layer1} (s)	t_{layer2}(s)		
		$vehicle1$	$vehicle2$	$vehicle3$
I	0.087	0.146	0.042	0.167
II	0.076	0.074	0.053	0.246

conflict zone. Besides, the spatial temporal corridors of different vehicles do not overlap in any time unit, ensuring safety during conflict resolution.

Comparing the results of ISTCs, the effect of different priority weights is distinct. A vehicle with higher priority weight tend to generate its spatial temporal corridor with a larger area, and the pivotal-points of corridor-cubes tend to be closer to the initial reference trajectory. Conversely, a vehicle with lower priority weight is more inclined to sacrifice their driving possibilities to make space for other high-priority vehicles. But anyway, regardless of whether the priority weight is larger or smaller, the corridor of each vehicle is always safe and satisfying dynamics, due to the constraints settled in Section IV. In group I, the spatial temporal corridor of $vehicle3$ was found to be the most aggressive in occupying the center of the intersection at the beginning of the conflict. While in group II, the center of the intersection was mostly occupied by the corridor of $vehicle1$. Furthermore, the results of the individual spatial temporal corridor with in-corridor trajectory are shown in Fig. 11, which also exposes the effect of priority weight. Comparing the corridor and trajectory of left-turning $vehicle3$ shown in Fig. 11(c) and (f), the results of group I were significantly better than those of group II, mainly in two aspects: 1) The overall space occupied by the corridor is greater. 2) The generated final trajectory is smoother. By contrast, when in the group II with $\eta^3 = 0.01$, the final trajectory of $vehicle3$ is highly restricted to the narrow corridor space.

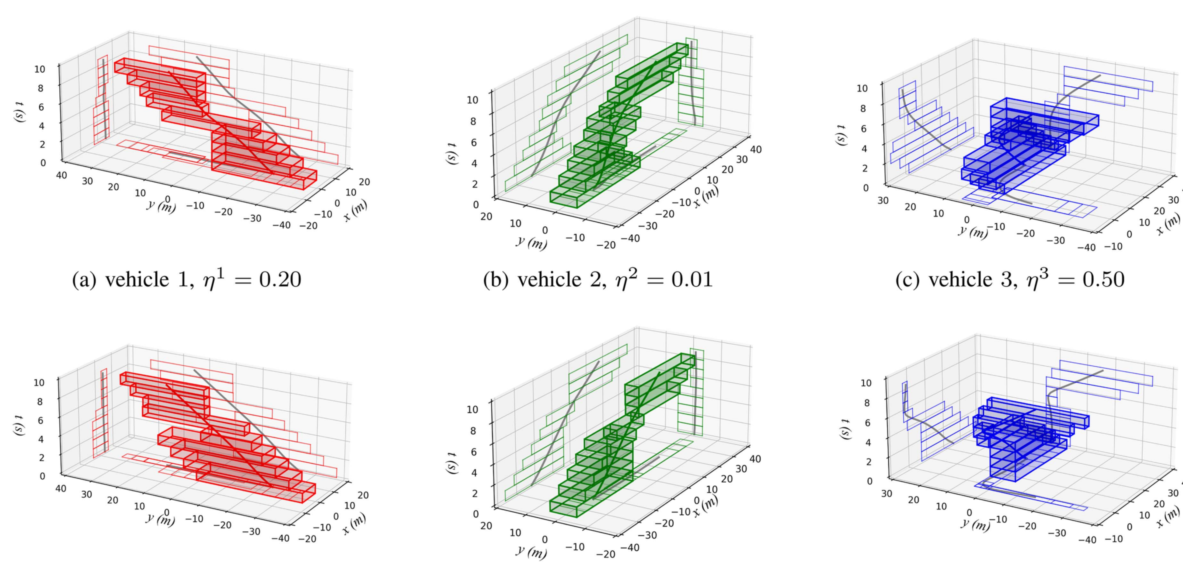
The multi-vehicle trajectory results as well as the velocity and acceleration results reflect the flexibility of the proposed method in the 3-D(x, y, t) space-time configuration space. These results can be obtained from the NLP solver statistics and (17) in Section IV, which are respectively shown in Figs. 12 and 13. In group I, $vehicle3$ with the highest priority weight performs a comfortably large radius left-turn under initial guidance. Meanwhile, $vehicle1$ and $vehicle2$ give way to $vehicle3$ mainly by adjusting their respective speeds, according to Fig. 13(a). By contrast, in group II, left-turning $vehicle3$ chose to make a detour to the core conflict area and turned at a smaller radius with lower comfort. While $vehicle1$ and $vehicle2$ benefited from this and gained more stable speeds, as shown in Fig. 13(a). Besides, it is demonstrated in both sets of experiments in Fig. 13 that all vehicles chose to slow down at the beginning of their involvement in the conflict. The magnitude of fluctuations in speed and acceleration is negatively correlated with the relative magnitude of the vehicle's priority weight.

The time consumption shown in Table III illustrates the efficiency of the proposed algorithm. Since the hierarchical framework is able to decouple multi-vehicle coordinated motion planning through ISTCs to the individual vehicle for trajectory



(a) ISTCs results of group I (b) ISTCs results of group II

Fig. 10. Comparison results of the multi-vehicle ISTCs. The spatial temporal corridors of $vehicle1$, $vehicle2$, and $vehicle3$ are shown in red, green, and blue, respectively. Inside each spatial temporal corridor are pivotal-points and their connecting lines.



(a) vehicle 1, $\eta^1 = 0.20$ (b) vehicle 2, $\eta^2 = 0.01$ (c) vehicle 3, $\eta^3 = 0.50$

(d) vehicle 1, $\eta^1 = 0.50$ (e) vehicle 2, $\eta^2 = 0.40$ (f) vehicle 3, $\eta^3 = 0.01$

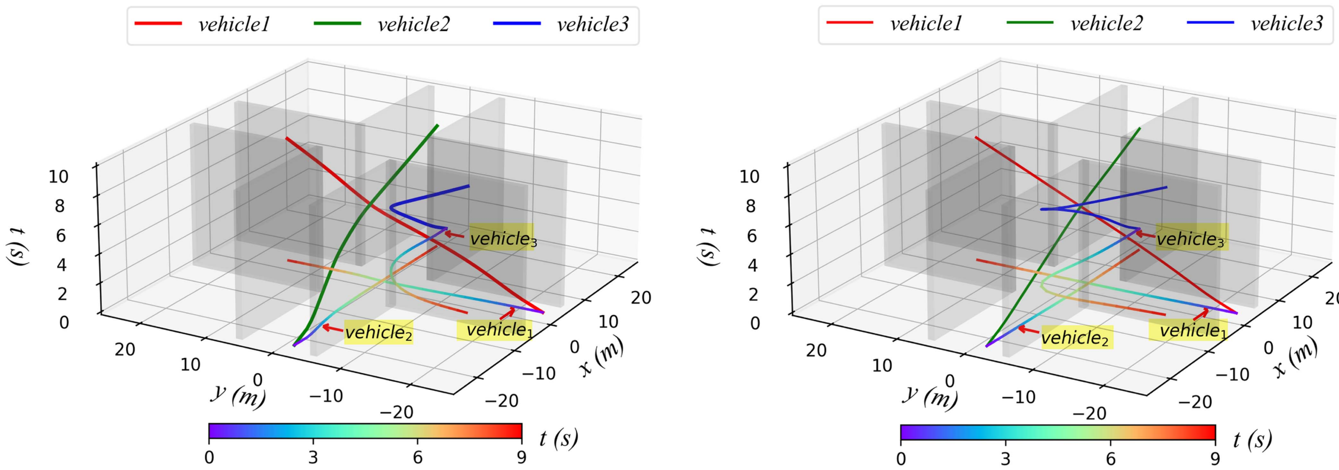
Fig. 11. The results of the individual spatial temporal corridor with the final trajectory inside; (a)–(c) are the results of group I, and (d)–(f) are the results of group II. 2-D projections of spatial temporal corridors and trajectories on each plane are also shown.

optimization, the time consumption can be viewed as two parts. In both experiment groups, the first layer of the framework was able to construct ISTCs rapidly. And in the second layer, each vehicle can obtain a safe and high-quality trajectory through the conflict area with an acceptable time consumption, fully considering vehicle kinematic characteristics. Finally, the distances between the final planned trajectories of the three vehicles shown in Fig. 14 reflect the safety properties of the proposed algorithm. In the two sets of simulations, the shortest distances between the planned trajectories are 2.44 m and 3.93 m, respectively.

To sum up, the experimental results in the unsignalized intersection are shown to be consistent with the original intention of the method detailed in Section IV. ISTCs can safely and efficiently resolve interaction conflicts and simplify the problem of generating collaborative trajectories. Flexible parameter settings allow vehicles to adopt different strategies when passing through conflict zones under the same environment and initial conditions. Besides, finding the solution in the 3D spatio-temporal configuration space greatly increases the flexibility of the vehicle to avoid conflicts.

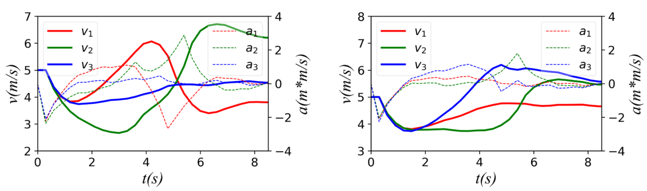
B. Comparison With Baseline Algorithms in the Challenging Dense Scenario

1) Description: In order to verify the feasibility and advantage of the proposed hierarchical method in general scenarios, we conducted comparison experiments with CL-CBS proposed in [13]. The experiments were carried out on a 50 m*38 m map in which 4 vehicles, 6 vehicles and 8 vehicles were tested with and without obstacles in the middle of the map. In order to



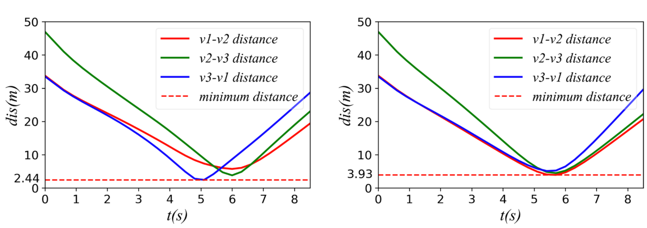
(a) Trajectory results of group I (b) Trajectory results of group II

Fig. 12. Comparison results with the final multi-vehicle trajectories. The gray translucent plane in the figure is the road boundary of the intersection. The trajectories of $vehicle1$, $vehicle2$, and $vehicle3$ in the 3-D (x, y, t) configuration space are drawn with red, green, and blue lines, respectively. In addition, the 2-D trajectory projection of the vehicles is also drawn on the $x - y$ plane, and the colorbar is used to represent the change of time.



(a) Group I (b) Group II

Fig. 13. Velocity and acceleration results in unsignalized intersection.



(a) Group I (b) Group II

Fig. 14. Distance between the trajectories of the three vehicles.

increase the challenge of resolving conflicts, the initial positions of vehicles were set on the left or right sides of the map, and all of them were required to move to the other side without collision. In our method, the initial guide trajectories are found using Hybrid A*, and a constant velocity of 5 m/s is set. In the simulation of both methods, the start and goal states are set to be the same.

In our method, when considering setting priorities for each vehicle, it makes more sense to differentiate priorities between vehicles with a strong competitive relationship. For example, in the eight-vehicle scenario shown in Fig. 15, changing the relative magnitude of the priority weights between $vehicle1$ and $vehicle2$ has a greater impact than changing it between

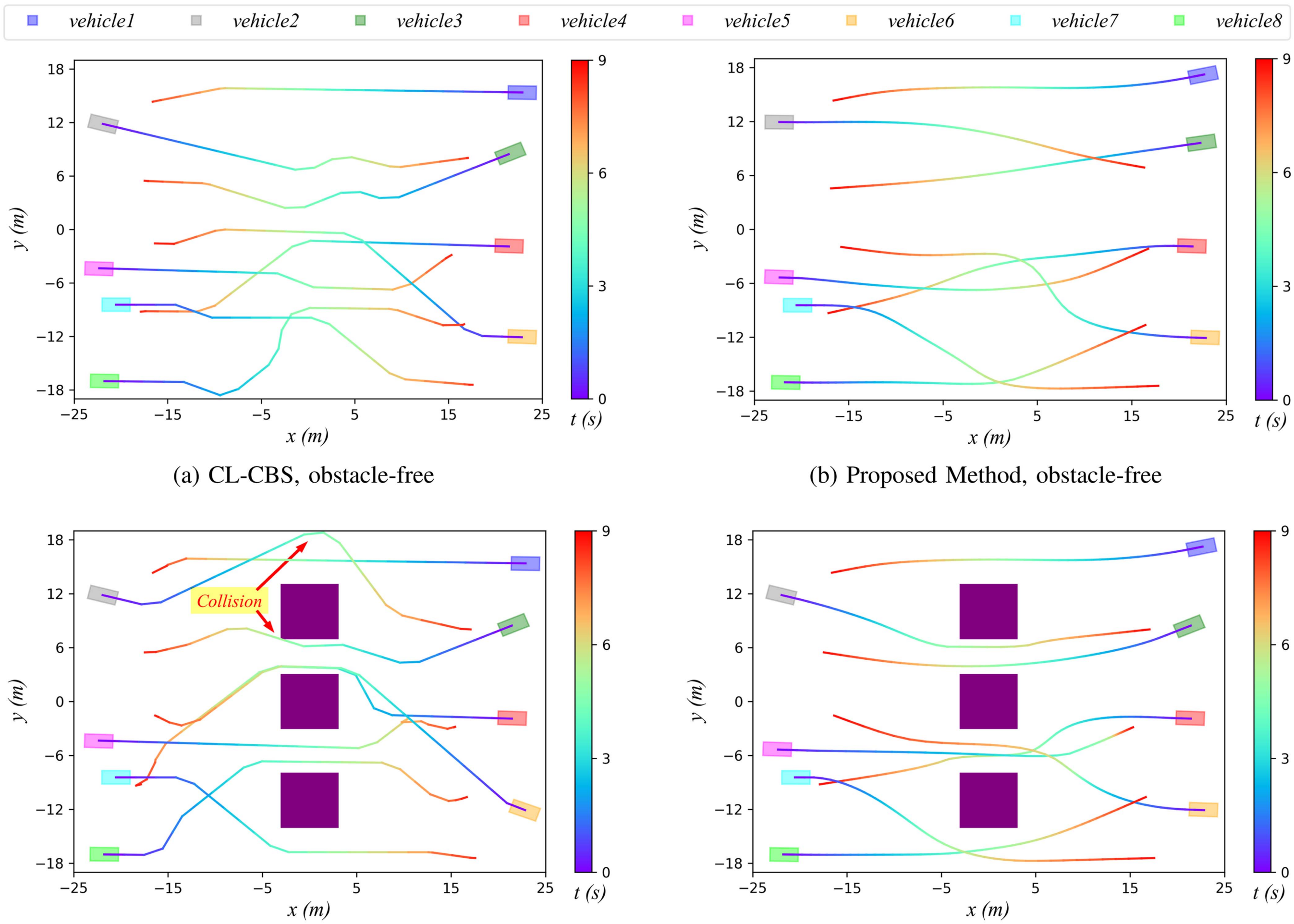
TABLE IV
PRIORITY WEIGHTS IN THE EIGHT-VEHICLE SCENARIO

Priority	η^1	η^2	η^3	η^4	η^5	η^6	η^7	η^8
Value	0.01	0.25	0.50	0.25	0.01	0.25	0.50	0.25

$vehicle1$ and $vehicle8$. On this basis, we focus on differentiating the priority weights of vehicles with competing relationships to increase the optimization gradient in the MIQP and improve the solution efficiency. In practice, we set the priority weights to three discrete values of 0.01, 0.25 and 0.50 as in the aforementioned unsignalized intersection simulation, and then differentially select the priority weights for vehicles with potential competition. For instance, the priority weights of each vehicle in the most challenging eight-vehicle scenario are taken as in Table IV.

The comparison results are shown in Table V. For CL-CBS, N_{node} is the number of conflict nodes that have been expended by conflict-based-search, t_{total} is the total time consumption of accomplishing the multi-vehicle planning task, L_{total} is the total length of the trajectories of all vehicles. For the proposed method, t_{l_1} is the time consumption of ISTCs construction in the first layer, t_{l_2} is the time consumption of trajectories optimization for all the vehicles in the second layer, t_{total} and L_{total} have the same meaning as in CL-CBS. Besides, the trajectories of 8 vehicles driving with and without obstacles are listed in Fig. 15.

2) Results and Discussion: The time consumption in both methods generally increases with the presence of obstacles and the number of vehicles, but our method is less sensitive to these factors and shows better performance in denser scenes. In the proposed framework, the first layer of MIQP was able to construct ISTCs for all participating vehicles efficiently in all simulations. Benefit from ISTCs decoupling the multi-vehicle collaboration problem, the computational complexity of the second layer for each individual vehicle is not affected by the



(a) CL-CBS, obstacle-free

(b) Proposed Method, obstacle-free

(c) CL-CBS, with obstacles

(d) Proposed Method, with obstacles

Fig. 15. Comparison results with Baseline Algorithms in the Challenging Dense Scenario. The purple rectangles indicate the position and shape of the obstacle, and the colorbar is used to indicate the temporal change of the trajectory. The locations of vehicle collision are marked by red arrows in (c).

TABLE V
COMPARISON RESULTS WITH BASELINE ALGORITHMS IN THE CHALLENGING DENSE SCENARIO

Vehicle numbers	Obstacles	CL-CBS			Proposed Method			
		N_{node}	$t_{total}(s)$	$L_{total}(m)$	$t_{l_1}(s)$	$t_{l_2}(s)$	$t_{total}(s)$	$L_{total}(m)$
4	×	5	0.069	175.554	0.171	0.402	0.573	157.615
4	✓	3	0.065	182.119	0.215	0.415	0.630	159.735
6	×	6	0.228	258.018	0.651	0.793	1.444	235.376
6	✓	225	9.611	272.611	1.622	0.774	2.436	239.282
8	×	530	22.98	352.066	3.954	0.977	4.931	313.634
8	✓	737	38.090	381.616	6.369	1.002	7.371	319.149

increasing complexity of the whole problem, so the total time consumption for the second layer is positively linearly related to the number of vehicles. In contrast, CL-CBS saw an explosive increase in the number of conflict nodes in denser scenes that were full of conflicts, resulting in a sharp escalation in time consumption. The reason for this phenomenon is that CL-CBS does not directly consider the collaboration between vehicles in the low-level planner, but first plans trajectories for each vehicle independently and then extends the conflict nodes by detecting conflicting positions. For each conflict node, CL-CBS also does not coordinate all the conflict-involved vehicles at the same time to resolve the conflict, but only applies a constraint to one vehicle at a time to make it bypass the conflict location.

Besides, in all the simulations, our method allowed vehicles to pass through the conflict zone with a relatively shorter path, because the ISTCs construction and trajectory optimization can adjust both the position and speed of vehicles in the 3D (x, y, t) configuration space, preserving a larger solution space. The results in Fig. 15 also demonstrate that our method can generate more delicate and smooth trajectories. As shown in Fig. 15(d), it can be found that ISTCs can make full use of the larger solution space to improve passage efficiency while ensuring vehicle safety. For example, the trajectories of $vehicle2$ and $vehicle3$ met in the narrow opening and passed through in parallel, while the trajectory of $vehicle5$ quickly passed through the nearest channel to make space for $vehicle4$ and $vehicle6$. By contrast in Fig. 15(c), the trajectories generated by CL-CBS was over-discretized, even leading to crossing the border or colliding with the obstacle, such as the trajectories of $vehicle2$ and $vehicle3$.

However, CL-CBS based on conflict-based-search is extremely efficient in finding feasible trajectories in the case of sufficient free space, much faster than the ISTCs-based method proposed in this article. The reason for this phenomenon is that the conflict-resolution mechanism of CL-CBS is to iteratively replan the trajectory of a single conflicting vehicle under the spatio-temporal constraint. In scenes with fewer vehicles and more free space, this mechanism has a low probability of causing new conflicts after resolving the current conflict, which greatly reduces the times of conflict node expansion and hybrid A* replanning. Differently, our method is to utilize a bi-level optimization of MIQP and NLP to globally coordinate all the vehicles simultaneously, so that takes longer to generate smoother trajectories.

VI. Conclusion

In this study, a novel hierarchical framework is proposed to accomplish multi-vehicle coordinated motion planning in general scenes. MIQP designed in the first layer can efficiently construct ISTCs, opening up safe spatial temporal corridors for each vehicle in complex interaction environments. The constraints imposed in the MIQP ensure the safety of vehicles while preserving enough margin for the feasible trajectory generation. Based on this, NLP settled in the second layer is capable of effectively generating safe and smooth in-corridor trajectories with the consideration of non-holonomic kinematics. Benefiting from the whole framework is optimized in the 3D (x, y, t) space-time configuration space, a larger solution space is preserved and the quality of the planning results is improved. Compared to other existing methods, the proposed bi-level method is able to decouple the global multi-vehicle motion planning task to individuals, simplifying the complexity and dispersing the computational pressure.

In future work, we will try to design more flexible spatial temporal corridor structures than cubes to further improve the adaptability and robustness of the algorithm.

References

[1] J. E. Siegel, D. C. Erb, and S. E. Sarma, "A survey of the connected vehicle landscape–architectures, enabling technologies, applications, and development areas," *IEEE Trans. Intell. Transp. Syst.*, vol. 19, no. 8, pp. 2391–2406, Aug. 2018.

[2] M. Hasan, S. Mohan, T. Shimizu, and H. Lu, "Securing vehicle-to-everything (V2X) communication platforms," *IEEE Trans. Intell. Veh.*, vol. 5, no. 4, pp. 693–713, Dec. 2020.

[3] J. Rios-Torres and A. A. Malikopoulos, "A survey on the coordination of connected and automated vehicles at intersections and merging at highway on-ramps," *IEEE Trans. Intell. Transp. Syst.*, vol. 18, no. 5, pp. 1066–1077, May 2017.

[4] J. Yu and S. M. LaValle, "Structure and intractability of optimal multi-robot path planning on graphs," in *Proc. AAAI Conf. Artif. Intell.*, 2013, pp. 1443–1449.

[5] D. Silver, "Cooperative pathfinding," in *Proc. AAAI Conf. Artif. Intell. Interactive Digit. Entertainment*, 2005, pp. 117–122.

[6] M. Barer, G. Sharon, R. Stern, and A. Felner, "Suboptimal variants of the conflict-based search algorithm for the multi-agent pathfinding problem," in *Proc. 7th Annu. Symp. Combinatorial Search*, 2014, pp. 961–962.

[7] D. Le and E. Plaku, "Multi-robot motion planning with dynamics via coordinated sampling-based expansion guided by multi-agent search," *IEEE Robot. Automat. Lett.*, vol. 4, no. 2, pp. 1868–1875, Apr. 2019.

[8] M. M. Khorshid, R. C. Holte, and N. R. Sturtevant, "A polynomial-time algorithm for non-optimal multi-agent pathfinding," in *Proc. 4th Annu. Symp. Combinatorial Search*, 2011, pp. 76–83.

[9] G. Wagner and H. Choset, "M*: A complete multirobot path planning algorithm with performance bounds," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2011, pp. 3260–3267.

[10] T. Standley, "Finding optimal solutions to cooperative pathfinding problems," in *Proc. AAAI Conf. Artif. Intell.*, 2010, pp. 173–178.

[11] M. Phillips and M. Likhachev, "SIPP: Safe interval path planning for dynamic environments," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2011, pp. 5628–5635.

[12] A. Andreychuk, K. Yakovlev, E. Boyarski, and R. Stern, "Improving continuous-time conflict based search," in *Proc. AAAI Conf. Artif. Intell.*, 2021, pp. 11220–11227.

[13] L. Wen, Y. Liu, and H. Li, "CL-MAPF: Multi-agent path finding for car-like robots with kinematic and spatiotemporal constraints," *Robot. Auton. Syst.*, vol. 150, 2022, Art. no. 103997.

[14] J. Li, M. Ran, and L. Xie, "Efficient trajectory planning for multiple non-holonomic mobile robots via prioritized trajectory optimization," *IEEE Robot. Automat. Lett.*, vol. 6, no. 2, pp. 405–412, Apr. 2021.

[15] R. Reijnen, Y. Zhang, W. Nuijten, C. Senaras, and M. Goldak-Altgassen, "Combining deep reinforcement learning with search heuristics for solving multi-agent path finding in segment-based layouts," in *Proc. IEEE Symp. Ser. Comput. Intell.*, 2020, pp. 2647–2654.

[16] M. Sinkar, M. Izhan, S. Nimkar, and S. Kurhade, "Multi-agent path finding using dynamic distributed deep learning model," in *Proc. IEEE Int. Conf. Commun. Inf. Comput. Technol.*, 2021, pp. 1–6.

[17] C. Bai, P. Yan, W. Pan, and J. Guo, "Learning-based multi-robot formation control with obstacle avoidance," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 8, pp. 11811–11822, Aug. 2022.

[18] M. Cai et al., "Multi-lane unsignalized intersection cooperation with flexible lane direction based on multi-vehicle formation control," *IEEE Trans. Veh. Technol.*, vol. 71, no. 6, pp. 5787–5798, Jun. 2022.

[19] W. Wang, C. Zhang, P. Wang, and C.-Y. Chan, "Learning representations for multi-vehicle spatiotemporal interactions with semi-stochastic potential fields," in *Proc. IEEE Intell. Veh. Symp.*, 2020, pp. 1935–1940.

[20] C. Xia, M. Xing, and S. He, "Interactive planning for autonomous driving in intersection scenarios without traffic signs," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 12, pp. 24818–24828, Dec. 2022.

[21] Z. Liang, J. Cao, S. Jiang, D. Saxena, and H. Xu, "Hierarchical reinforcement learning with opponent modeling for distributed multi-agent cooperation," in *Proc. IEEE 42nd Int. Conf. Distrib. Comput. Syst.*, 2022, pp. 884–894.

[22] B. Li, Y. Zhang, Y. Feng, Y. Zhang, Y. Ge, and Z. Shao, "Balancing computation speed and quality: A decentralized motion planning method for cooperative lane changes of connected and automated vehicles," *IEEE Trans. Intell. Veh.*, vol. 3, no. 3, pp. 340–350, Sep. 2018.

[23] S. Xie, J. Hu, P. Bhowmick, Z. Ding, and F. Arvin, "Distributed motion planning for safe autonomous vehicle overtaking via artificial potential field," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 11, pp. 21531–21547, Nov. 2022.

[24] R. Deng, B. Di, and L. Song, "Cooperative collision avoidance for overtaking maneuvers in cellular V2X-based autonomous driving," *IEEE Trans. Veh. Technol.*, vol. 68, no. 5, pp. 4434–4446, May 2019.

[25] Z. e. a. Kherroubi, S. Aknine, and R. Bacha, "Novel decision-making strategy for connected and autonomous vehicles in highway on-ramp merging," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 8, pp. 12490–12502, Aug. 2022.

[26] H. Liu, W. Zhuang, G. Yin, R. Li, C. Liu, and S. Zhou, "Decentralized on-ramp merging control of connected and automated vehicles in the mixed traffic using control barrier functions," in *Proc. IEEE Int. Intell. Transp. Syst. Conf.*, 2021, pp. 1125–1131.

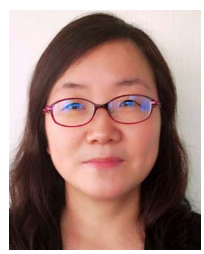
[27] A. Nakamura, Y.-C. Liu, and B. Kim, "Short-term multi-vehicle trajectory planning for collision avoidance," *IEEE Trans. Veh. Technol.*, vol. 69, no. 9, pp. 9253–9264, Sep. 2020.

[28] B. Wang, Y. Lu, T. Zhang, J. Gong, and H. Zhao, "Multi-vehicle coordinated motion planning based on interactive primitive tree," in *Proc. IEEE Int. Conf. Unmanned Syst.*, 2021, pp. 51–57.
[29] T. Kessler and A. Knoll, "Cooperative multi-vehicle behavior coordination for autonomous driving," in *Proc. IEEE Intell. Veh. Symp.*, 2019, pp. 1953–1960.
[30] X. Shen and F. Borrelli, "Multi-vehicle conflict resolution in highly constrained spaces by merging optimal control and reinforcement learning," 2022, *arXiv:2211.01487*.
[31] J. Huang and C. Liu, "Multi-car convex feasible set algorithm in trajectory planning," in *Proc. Dyn. Syst. Control Conf.*, 2020, Art. no. V001T10A003.
[32] H. Zhou and C. Liu, "Distributed motion coordination using convex feasible set based model predictive control," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2021, pp. 8330–8336.
[33] R. Deits and R. Tedrake, "Computing large convex regions of obstacle-free space through semidefinite programming," in *Algorithmic Foundations of Robotics XI: Selected Contributions of the Eleventh International Workshop on the Algorithmic Foundations of Robotics*. Berlin, Germany: Springer, 2015, pp. 109–124.
[34] S. Liu et al., "Planning dynamically feasible trajectories for quadrotors using safe flight corridors in 3-D complex environments," *IEEE Robot. Automat. Lett.*, vol. 2, no. 3, pp. 1688–1695, Jul. 2017.
[35] W. Ding, L. Zhang, J. Chen, and S. Shen, "Safe trajectory generation for complex urban environments using spatio-temporal semantic corridor," *IEEE Robot. Automat. Lett.*, vol. 4, no. 3, pp. 2997–3004, Jul. 2019.
[36] T. Zhang, M. Fu, W. Song, Y. Yang, and M. Wang, "Trajectory planning based on spatio-temporal map with collision avoidance guaranteed by safety strip," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 2, pp. 1030–1043, Feb. 2020.
[37] T. Zhang, W. Song, M. Fu, Y. Yang, X. Tian, and M. Wang, "A unified framework integrating decision making and trajectory planning based on spatio-temporal voxels for highway autonomous driving," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 8, pp. 10365–10379, Aug. 2022.
[38] S. Deolasee, Q. Lin, J. Li, and J. M. Dolan, "Spatio-temporal motion planning for autonomous vehicles with trapezoidal prism corridors and Bézier curves," 2022, *arXiv:2209.15150*.
[39] J. Li, X. Xie, Q. Lin, J. He, and J. M. Dolan, "Motion planning by search in derivative space and convex optimization with enlarged solution space," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2022, pp. 13500–13507.
[40] M. Morsali, E. Frisk, and J. Åslund, "Spatio-temporal planning in multi-vehicle scenarios for autonomous vehicle using support vector machines," *IEEE Trans. Intell. Veh.*, vol. 6, no. 4, pp. 611–621, Dec. 2021.
[41] D. Dolgov, S. Thrun, M. Montemerlo, and J. Diebel, "Practical search techniques in path planning for autonomous driving," *Ann Arbor*, vol. 1001, no. 48105, pp. 18–80, 2008.
[42] M. Berkelaar, K. Eikland, and P. Notebaert, "lpsolve: Open source (mixed-integer) linear programming system," Eindhoven Univ. Technol., Eindhoven, The Netherlands, vol. 63, 2004.
[43] L. Gurobi Optimization, "Gurobi optimizer reference manual," 2023. [Online]. Available: https://www.gurobi.com
[44] L. T. Biegler and V. M. Zavala, "Large-scale nonlinear programming using IPOPT: An integrating framework for enterprise-wide dynamic optimization," *Comput. Chem. Eng.*, vol. 33, no. 3, pp. 575–582, 2009.

Xiang Zhang received the B.S. degree in vehicle engineering in 2022 from Beijing Institute of Technology, Beijing, China, where he is currently working toward the M.S. degree with the Intelligent Vehicle Research Center, School of Mechanical Engineering. His research interests include intelligent vehicles, motion planning and control.

Boyang Wang received the B.S degree in vehicle engineering and the Ph.D. degree from the Beijing Institute of Technology, Beijing, China, in 2013 and 2020, respectively. From 2017 to 2019, he was a Visiting Researcher with the Interaction Digital Human Group, CNRS-UM LIRMM. From 2020 to 2022, he was a Postdoctoral Researcher with the Key Laboratory of Machine Perception (MOE), Peking University, Beijing. He is currently an Assistant Professor with the School of Mechanical Engineering, Beijing Institute of Technology. His research interests include intelligent vehicles, driver behavior, motion planning and control, and vehicle dynamics modeling.

Yaomin Lu received the B.S degree in vehicle engineering from Hunan University, Changsha, China and M.S. degree in mechanical engineering from the Beijing Institute of Technology, Beijing, China, in 2019 and 2022, respectively. She is currently with Baidu Intelligent Driving Group, Beijing, China. Her research interests include intelligent vehicles, motion planning, and optimization method.



Haiou Liu received the B.S. and Ph.D. degrees from the Beijing Institute of Technology, Beijing, China, in 1998 and 2003, respectively. From 2013 to 2014, she was a Visiting Scholar with the Energy and Automotive Research Laboratory, Mechanical Engineering Department, Michigan State University, East Lansing, MI, USA. She is currently a Professor with the School of Mechanical Engineering, Beijing Institute of Technology. Her teaching interests focuses on vehicle control classes at both undergraduate and graduate levels. Her research interests include design and control of automated manual transmission and hybrid powertrain.

Jianwei Gong received the B.S. degree from the National University of Defense Technology, Changsha, China, in 1992, and the Ph.D. degree from Beijing Institute of Technology, Beijing, China, in 2002. Between 2011 and 2012, he was a Visiting Researcher of Robotic Mobility Group, Massachusetts Institute of Technology, Cambridge, MA, USA. He is currently a Professor and the Director of the Intelligent Vehicle Research Center, School of Mechanical Engineering, Beijing Institute of Technology. His research interests include intelligent vehicle environment perception and understanding, driver behavior, motion planning and control.

Huiyan Chen received the Ph.D. degree from Beijing Institute of Technology, Beijing, China, in 2004. Since 1981, he has been with Beijing Institute of Technology. He was the Director of the Intelligent Vehicle Research Center. He is currently a Professor with the School of Mechanical Engineering, Beijing Institute of Technology. His research interests include intelligent vehicles, powertrain system modeling an control, and information technologies.